

CHANDNI VERMA

📍 Punjab, India ✉ chandniverma.geo@gmail.com 📄 Portfolio 🌐 GitHub 🔗 LinkedIn

👤 PROFILE

Remote Sensing and Geospatial professional with experience in geospatial analysis, satellite image processing, SAR/InSAR, and Python-based geospatial automation. Skilled in developing reproducible workflows to transform Earth observation data into practical insights.

🏢 PROFESSIONAL EXPERIENCE

Junior Research Fellow 09/2024 – 03/2026 | Chandigarh, India
CSIR-Central Scientific Instruments Organisation (CSIO)

A comprehensive Geomorphic Anomaly Investigation over Sutlej Basin, India

- Conducted geomorphic, hydrological, and morphotectonic analyses using multi-source geospatial datasets and computational tools including Python, MATLAB, TopoToolbox, and GIS platforms.
- Independently developed an automated Python-based tool for topographic line and swath profile extraction, enabling scalable terrain analysis and publication-quality vector outputs.
- Performed watershed and drainage-network delineation, fluvial terrace mapping, longitudinal river-profile analysis, basin asymmetry assessment, and χ -index analysis to investigate landscape evolution and tectonic influences in the Himalayan terrain.
- Conducted field investigations across Himachal Pradesh to validate geomorphic interpretations, including terrace verification, elevation measurements, lithological logging, sediment characterization, geomorphic documentation, and soil-sample collection for OSL dating.
- Prepared geomorphic maps, technical figures, visualizations, project reports, and research documentation.

Trainee - Advanced SAR Data Processing for Ground Surface Movement 10/2024 – 10/2024 | Dehradun, India

Indian Institute of Remote Sensing (IIRS), ISRO

- Configured Linux-based computational environments and preprocessing workflows for SAR interferometric analysis.
- Performed guided Differential SAR Interferometry (DInSAR) and Small Baseline Subset (SBAS) processing workflows using ASF, SNAP, and ISCE for ground-surface deformation analysis.
- Acquired practical exposure to SAR time-series analysis frameworks, including MintPy, for deformation monitoring and interferometric interpretation.

🎓 EDUCATION

M.Tech in Remote Sensing and GIS (Geosciences) 08/2022 – 07/2024 | Dehradun, India
Indian Institute of Remote Sensing (IIRS), ISRO

M.Sc in Geology 07/2014 – 06/2016 | Chandigarh, India
Panjab University

B.Sc in Geology 07/2011 – 06/2014 | Chandigarh, India
Panjab University
Minors: Mathematics and Chemistry

📁 PROJECTS

GeoProfiler / GeoProfilerX: Automated Raster Profile Extraction and Geospatial Analysis

Python | Streamlit | Rasterio | GeoPandas | NumPy | SciPy | Matplotlib

- Developed **GeoProfiler**, an open-source Python tool for automated extraction of topographic line and swath profiles from DEMs, with terrain visualization and quantitative profile analysis.
- Extended GeoProfiler into an interactive Streamlit web application, providing accessible browser-based DEM visualization, profile generation, spatial-data handling, and export of analysis results.
- Subsequently developed **GeoProfilerX**, generalizing the workflow from DEMs to projected single-band raster datasets, including DEMs, gravity/magnetic anomalies, and InSAR displacement data.
- Implemented automated raster sampling, profile generation, smoothing, visualization, and data export, creating a reusable workflow for quantitative analysis of diverse geoscientific raster datasets.
- The project demonstrates experience in scientific programming, geospatial data processing, workflow automation, visualization, and reproducible computational tool development.

Web App: geoprofiler.streamlit.app 🌐

GitHub: <https://github.com/chandnivermageo/GeoProfilerX> 📄

Land Subsidence Characterization in Northwest India in relation to Groundwater Overextraction, Hydrogeology, and Active Tectonics

DInSAR | PS-InSAR | Sentinel-1 | GRACE | Aquifer System Compaction

- Analyzed GRACE-derived groundwater-storage anomalies to investigate regional groundwater-depletion trends.

- Processed selected Sentinel-1 SLC image pairs using DInSAR workflows in ENVI/SARscape and performed PS-InSAR time-series analysis of 151 Sentinel-1 SLC scenes (2018–2022) in SARPROZ to derive LOS displacement time series.
- Modeled aquifer-system compaction using piezometric groundwater-level data and compared the modeled deformation with PS-InSAR-derived displacement trends.
- Conducted field investigations at subsidence-affected locations to assess deformation-related impacts through on-site observations and photographic documentation of structural damage to buildings.

SAR-Based Reservoir Sedimentation and Storage Capacity Assessment using Sentinel-1 GRD Imagery

Sentinel-1 | Google Earth Engine

- Conducted SAR-based reservoir sedimentation assessment using Sentinel-1 GRD imagery and multi-temporal water-spread analysis in Google Earth Engine (GEE) and ArcGIS.
- Integrated SAR-derived water-spread extent with India WRIS reservoir-level data to estimate revised storage capacity and sedimentation impacts.

Snow-Cover Delineation and Change Detection using Landsat 8 and DEM Integration

Python | Landsat 8 | DEM

- Developed an automated Python-based workflow for snow-cover delineation and change detection using a multi-criteria classification approach integrating spectral indices, radiometric thresholds, elevation filtering, and morphological cleanup for improved snow extraction accuracy.
- Performed raster-based spatial analysis, binary change mapping, and elevation-stratified snow distribution assessment using Rasterio, NumPy, SciPy, and Matplotlib.

GitHub: github.com/chandnivermageo/Snow-Cover-Change-Detection 

TECHNICAL SKILLS

Remote Sensing & Earth Observation: Optical and SAR image processing, DInSAR, PS-InSAR, SAR time-series analysis, change detection, snow and water-body delineation.

Terrain & Geomorphic Analysis: DEM processing, terrain analysis, watershed and drainage delineation, topographic profile extraction, fluvial geomorphology, terrace mapping, morphotectonic analysis.

Programming & Scientific Computing: Python, Streamlit, MATLAB, R, Rasterio, NumPy, SciPy, GeoPandas, Matplotlib, TopoToolbox, geospatial workflow automation.

SOFTWARE & TOOLS

ArcGIS • QGIS • Google Earth Engine • ENVI/SARscape • SNAP • SARPROZ • ERDAS IMAGINE • Google Earth Pro

WORKSHOP & CONFERENCES

- National Aquifer Mapping (NAQUIM) and Groundwater Data Dissemination for Sustainable Management, CGWB. Workshop, March 2026.
- Observing Ground Deformation from Space and in the Field, CSIR-CSIO. International Workshop, December 2024.
- IIRS Academia Meet (IAM-2024). Poster presentation, *Land Subsidence Characterization in Northwest India using GRACE and SAR Data*, 2024.
- 6th World Congress on Disaster Management (2023). Abstract selected, *Reservoir Sedimentation Assessment of Tehri Reservoir Using SAR Data*.
- 10th Chandigarh Science Congress (CHASCON-2016). Poster presentation, 2016.

LANGUAGES

Hindi, English, and Punjabi

ACHIEVEMENTS AND SCHOLARSHIP

- GATE 2025 (GE-Geomatics): All India Rank 153, GATE Score 496
- GATE 2024 (GE-Geomatics): All India Rank 239, GATE Score 445
- Recipient AICTE M.Tech GATE Scholarship (2022-2024)
- GATE 2022 (GG-Geology & Geophysics): All India Rank 763, GATE Score 370

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- Winner (Mixed Table Tennis & Badminton), CSIR-CSIO Sports Tournament, 2025
- Key team member in organizing the International Conference on Land Surface Deformation at CSIR-CSIO (Dec 2024)
- Held leadership roles as General Secretary (2015-16) and Cultural Secretary (2013-14) at Pick & Hammer Club, Panjab University

INTERESTS

Singing, Guitar, Badminton and Table Tennis